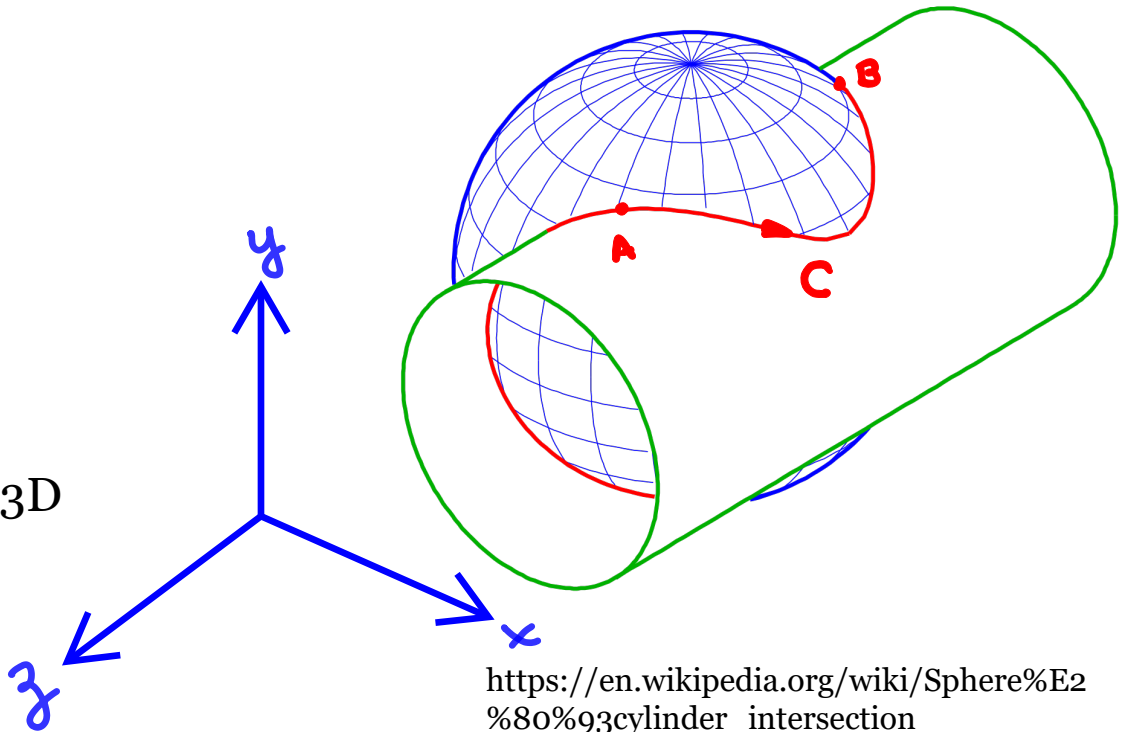


Path and Curves

Trim Ch 11, S. 11.10

Outline:

1. Path and curves in 3D
2. Graphing and orienting 3D curves



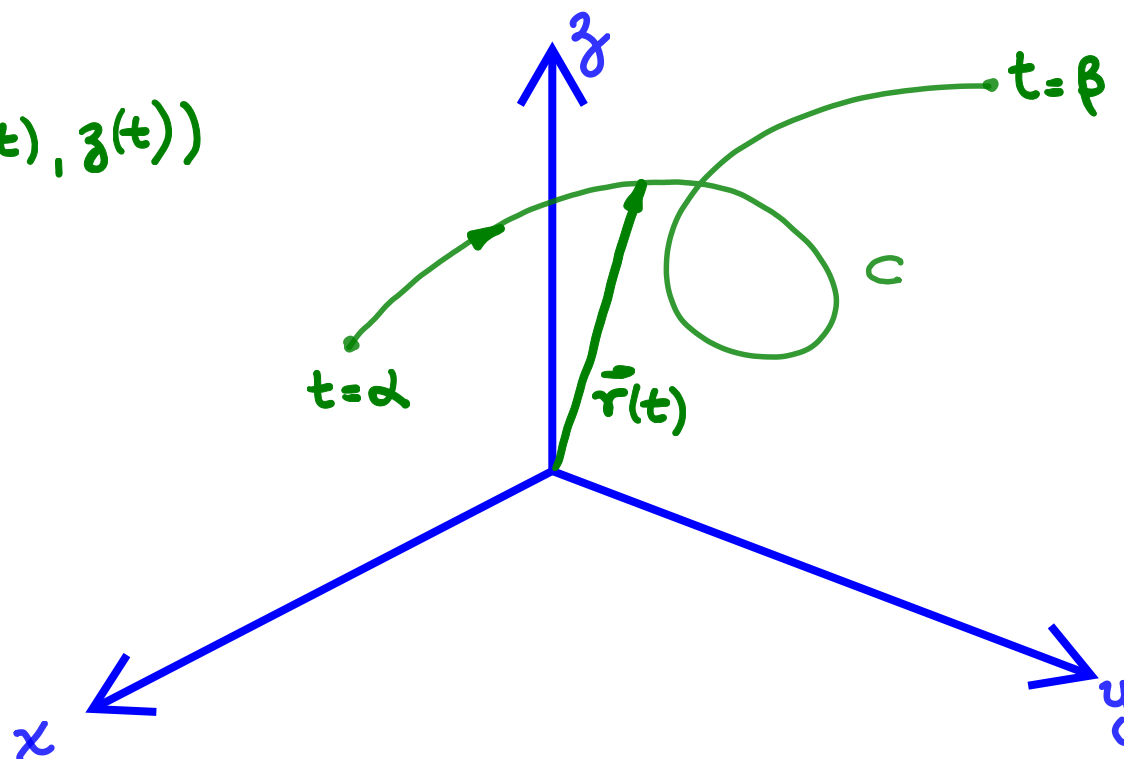
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Path and Curves

Let us put together what we already know about vector functions,

$$C: \vec{r}(t) = (x(t), y(t), z(t))$$

$$\alpha \leq t \leq \beta$$



Orientation:

The direction of a path always corresponds to the direction in which the parameter “t” increases along the path.

Paths and Curves

In summary ...

Path:

A path in $\mathbb{R}^3$ (or $\mathbb{R}^2$) is a **VECTOR FUNCTION**

$$\vec{r}(t) = (x(t), y(t), z(t))$$

where $x(t)$, $y(t)$ and $z(t)$ are called component functions, and the interval of the parameter t ($\alpha \leq t \leq \beta$) parametrizes the curve C .

Curve:

The **IMAGE** or drawing of a path $\vec{r}(t)$ is called a **curve** in $\mathbb{R}^3$ (or $\mathbb{R}^2$)

The vector function $\vec{r}(t)$ is known as a **PATH or a **PARAMETRIZATION** of the curve C**

Graphing and Orienting a Curve

Example:

Graph the curve given by the following path and indicate its direction.

$$\vec{r}(t) = (\cos t, \sin t, t), \quad 0 \leq t < \infty$$

Solution:

DOMAIN: THE THREE COMPONENTS ARE DEFINED FOR ALL t 'S

GRAPH/CURVE: COMPONENT FUNCTIONS ARE

$$x(t) = \cos t$$

$$y(t) = \sin t$$

$$z(t) = t$$

FORGETTING "z" FOR A SEC:
 $x^2 + y^2 = \cos^2 t + \sin^2 t = 1$

CIRCLE IN X-Y
PLANE

$$x^2 + y^2 = 1 \rightarrow \text{EQ' OF CYLINDER}$$

→ THE CURVE LIES ON A CYLINDER !

Graphing and Orienting a Curve

Let us evaluate some position vectors in the path $\vec{r}(t) = (\cos t, \sin t, t)$,

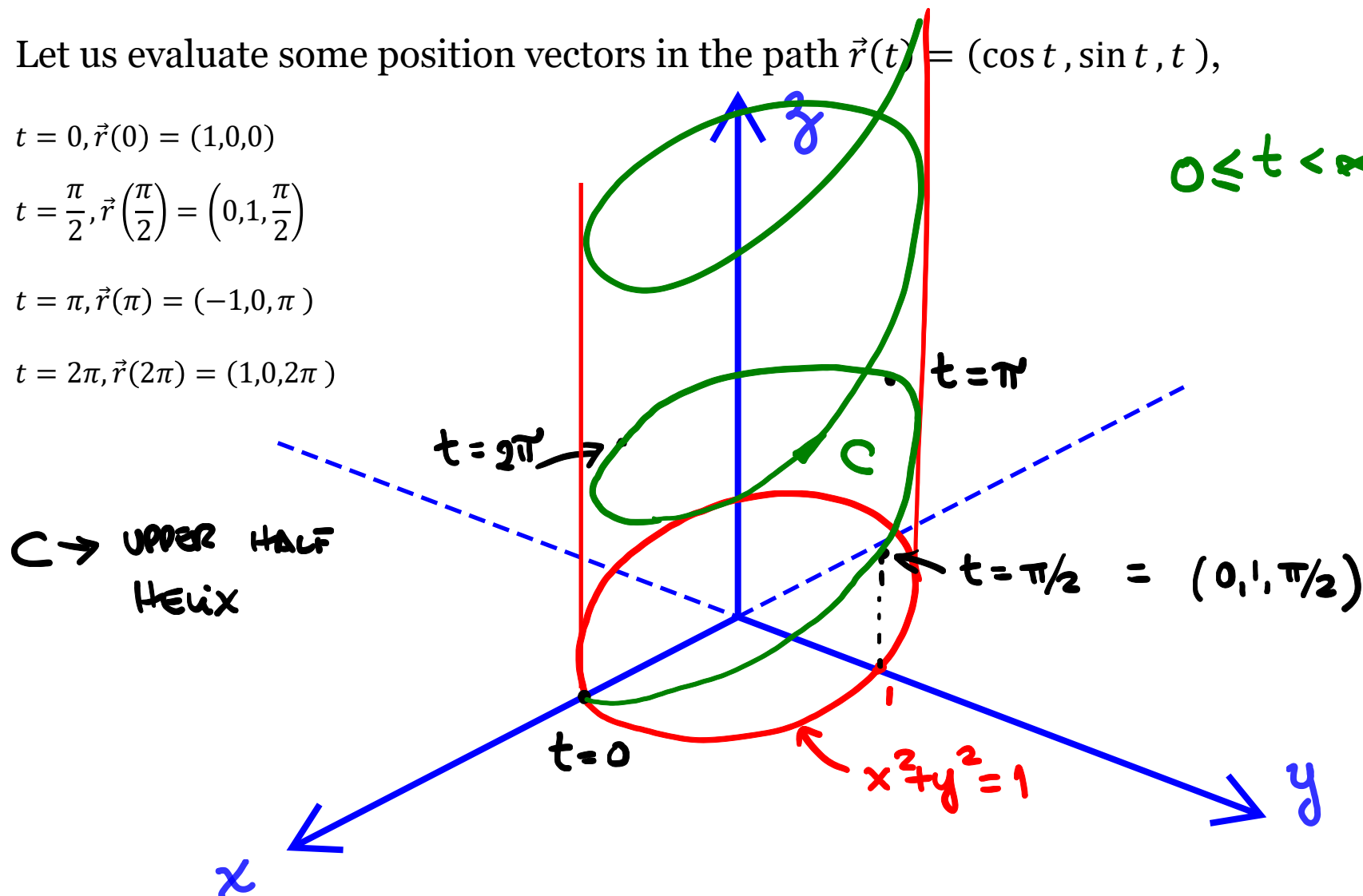
$$t = 0, \vec{r}(0) = (1, 0, 0)$$

$$t = \frac{\pi}{2}, \vec{r}\left(\frac{\pi}{2}\right) = \left(0, 1, \frac{\pi}{2}\right)$$

$$t = \pi, \vec{r}(\pi) = (-1, 0, \pi)$$

$$t = 2\pi, \vec{r}(2\pi) = (1, 0, 2\pi)$$

$$0 \leq t < \infty$$



Graphing and Orienting a Curve

Some of the questions we can ask here are:

1. What if instead, we have $\vec{r}(t) = (\cos t, t, \sin t)$

CYLINDER OPENS IN y-AXIS, AND HELIX ROTATES ABOUT THAT AXIS

2. What if we allow for different coefficients such as $\vec{r}(t) = (9\cos t, 2\sin t, t)$

CROSS SECTION OF CYLINDER WILL BE AN ELLIPSE

3. What happens if instead of $z = t$, we have $z = t^2$, that is $\vec{r}(t) = (\cos t, \sin t, t^2)$

THE HELIX WILL BE RISING UP THE CYLINDER FASTER !

Problem

- a. Demonstrate that the curve given by,

$$\vec{r}(t) = \sin t \hat{i} + \cos t \hat{j} + \sin^2 t \hat{k}$$

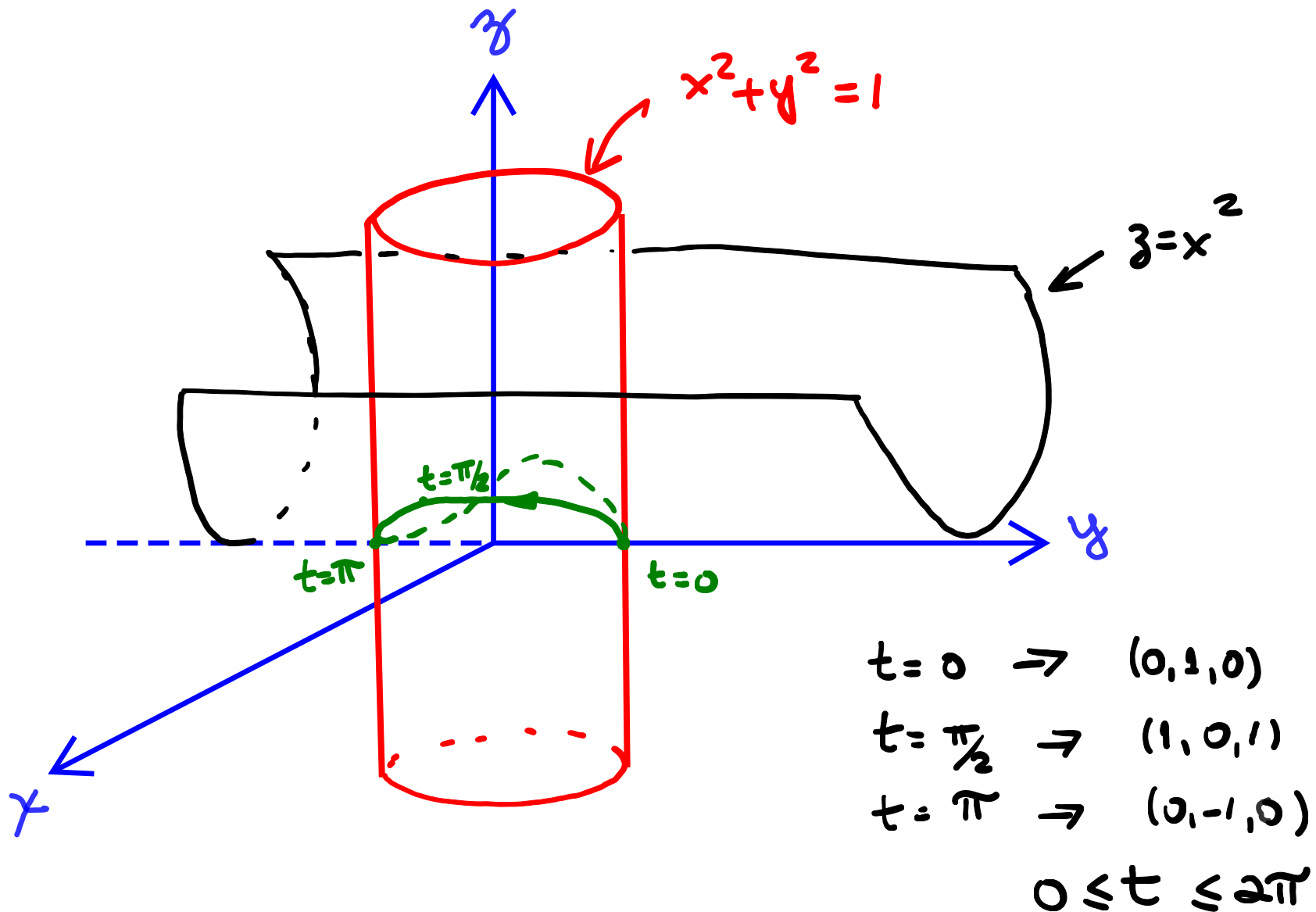
is formed by the intersection of two surfaces.

- b. Use this fact to help sketch the curve
- c. Draw the direction of the curve and determine the range of the parameter t

a) IF $\begin{cases} x = \sin t \\ y = \cos t \end{cases} \Rightarrow x^2 + y^2 = 1 \rightarrow \text{CYLINDER}$

WE ALSO NOTICE $\Rightarrow x^2 = z \rightarrow \text{PARABOLIC CYLINDER}$

Problem



Summary

- The vector function $\vec{r}(t)$ is known as a **PATH** or a **PARAMETRIZATION** of the curve C
- To graph a path, we relate the component functions within their domains in the rectangular coordinate system
- We plot some points to help visualization
- To determine the direction of a curve within the parameter range, we need to plot a minimum of three points on the curve



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Thank you